A proof of the conjectured order of the recurrence for OEIS A223972

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1 The conjecture

The Online Encyclopedia of Integer Sequences records A223972 as

Number of nX5 0..2 arrays with rows and antidiagonals unimodal

and carries in its Formula section the line:

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Empirical recurrence of order 43 (see link above)
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That line carries no separate signature; the entry itself is by R. H. Hardin (Mar 30 2013).

The word *Empirical* — equivalently *Conjecture* — is the entry's own: the statement is recorded as fitted to the published terms, and no proof is given there. As of revision 10, last modified Oct 03 2025, the entry records no proof and links to none, so the statement is open. This note settles it.

The entry states only that the sequence satisfies a linear recurrence of order 43 and refers to a linked file for its coefficients, which this note does not use. What is proved below is the corresponding exact statement: the least order of a linear recurrence the sequence eventually satisfies, and the exact index beyond which it holds.

2 The object

The name is read as follows. The reading is not a guess: it is pinned against the entry's own published terms, and against an independent enumeration, before anything is proved (Section 7).

Let $q = 3$ and let A be an $n' \times 5$ array over $\{0, \ldots, 2\}$ with $n' = n$ rows.

A row is read left to right, a column top to bottom, a diagonal downwards and to the right, an antidiagonal downwards and to the left. A sequence is *unimodal* when it is nondecreasing and then nonincreasing.

Every one of the array's antidiagonals is unimodal.

Every one of the array's rows is unimodal.

$a(n)$ is the number of such arrays. The entry's offset is 1, so its term of index n is the count for n rows.

3 The count is a walk count

Read the array one row at a time. Conditions along a row are decided by that row alone. A condition along a column, a diagonal or an antidiagonal compares entries in consecutive rows, so it is decided one step at a time — except that unimodality along such a line must remember whether that line has already turned from rising to falling.

Accordingly the state is the last row together with one bit for each of the antidiagonals through its entries, saying whether that line has already turned. Adding a row moves those bits along the line: a column keeps its own bit, a diagonal passes its bit one place to the right and starts a fresh line at the left border, an antidiagonal passes it one place to the left and starts a fresh line at the right border.

An array of n' rows is then exactly a walk of length n' from the empty state, and every array arises from exactly one walk. With M the adjacency matrix of the state digraph, w the indicator of its start and f of its accepting states, $a(n) = w^{\mathsf{T}} M^{\,n'} f$, so the count is C-finite.

4 The degree bound, derived

The reachable part of that digraph has 696 states, 696 after discarding states from which no accepting state is reachable. States from which the same number of arrays can be completed, for every remaining number of rows, are then identified by partition refinement — start from the partition by acceptance and separate two states as soon as they send different numbers of edges into some block. On the blocks the number of completions of each length is constant, so the quotient counts exactly what the original does.

The refinement, applied first forwards and then in the mirror direction on the edges coming in, leaves $S = 85$ blocks, and the count satisfies the linear recurrence given by the characteristic polynomial of the 85×85 quotient matrix. That degree bound is computed here, not assumed, and it is the only one the decision procedure below uses.

5 The decision procedure

Let $c_1, \ldots, c_D$ be the coefficients of a claimed recurrence $a(n) = \sum_{i=1}^{D} c_i\, a(n-i)$, let $q(t) = t^D - \sum_{i=1}^{D} c_i t^{D-i}$ be its characteristic polynomial, and put

$$u_m \;=\; \tilde{w}^{\mathsf{T}} L^{\,m-1} q(L)\, \tilde{f} \qquad (m \geq 1).$$

Expanding the powers of L and using the displayed formula for a, one gets $u_m = a(n) - \sum_{i=1}^{D} c_i\, a(n-i)$ with $n = m + D$. So u_m is exactly the residual of the claim at that index, and the recurrence holds at an index n if and only if the corresponding u_m vanishes.

Now $u_m = \tilde{w}^{\mathsf{T}} L^{m-1} y$ with $y = q(L)\tilde{f}$ a fixed vector, so (u_m) satisfies the characteristic polynomial of L, of degree 85: writing that polynomial as $t^{85} - \sum_{i=1}^{85} \lambda_i t^{85-i}$ gives $u_{m+85} = \sum_{i=1}^{85} \lambda_i u_{m+85-i}$. Hence 85 consecutive vanishing residuals force every later residual to vanish, by induction. The test is therefore finite; and because it locates the *last* nonvanishing residual, it pins the threshold exactly rather than merely bounding it.

Everything is carried out in exact integer arithmetic on the vector $L^{m-1}y$. The matrix M is never formed.

6 Results

Theorem 1. *For A223972, the least order of a linear recurrence that a eventually satisfies is exactly* 43*, and that recurrence holds for every* $n > 43$*, the entry's own figure.*

Proof. The count is C-finite of order at most $S = 85$, so the minimal annihilator of the tail $(a(n))_{n \geq M}$ is the same for every M beyond S: the nilpotent part of L is killed after S steps. Taking such an M and 176 consecutive terms from there, the Berlekamp–Massey algorithm over $\mathbb{Q}$ returns the minimal linear recurrence generating them; since the sequence satisfies one of order at most S, $2S$ terms determine it. Its order is 43. That recurrence was then put through the residual test of Section 5: every residual vanishes, followed by more than $S = 85$ consecutive vanishing ones, so it holds for every larger n. No recurrence of smaller order can hold eventually, since the tail's minimal annihilator is what the algorithm returned. □

7 Verification

Four checks were run, each reproducible from the code accompanying this note, and the first two are what pin the reading of the entry's English.

- **The published terms.** The walk count reproduces all 13 terms the entry publishes, exactly, in integer arithmetic, with the index taken from the entry's stated offset (1) rather than fitted. The first of them are 86, 7396, 440628, 22935921, 1140963027, 55803232969, 2708281019793, 131014406127439, ...
- **An independent enumeration.** For $n = 1, \ldots, 2$ every one of the 3^{5n} arrays was written out and tested cell by cell against the condition as the entry states it — no transfer matrix, no row window, a separate piece of code. The counts agree with the walk count and with the entry.
- **The claim on the raw data.** Each claim was evaluated on the entry's published terms alone, with no model involved, wherever the proved range and the published data overlap. It holds there.
- **The annihilation test.** Carried to the derived bound $S = 85$ over the whole vector, in exact integer arithmetic, never sampled and never truncated.

8 References

1. OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, entry A223972, revision 10, last modified Oct 03 2025.

2. R. P. Stanley, *Enumerative Combinatorics*, Vol. 1, 2nd ed., Cambridge University Press, 2012; §4.7, the transfer-matrix method.

3. M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011; C-finite sequences and their closure properties.

4. J. Berstel and C. Reutenauer, *Noncommutative Rational Series with Applications*, Cambridge University Press, 2011; linear representations of counting automata and their reduction.